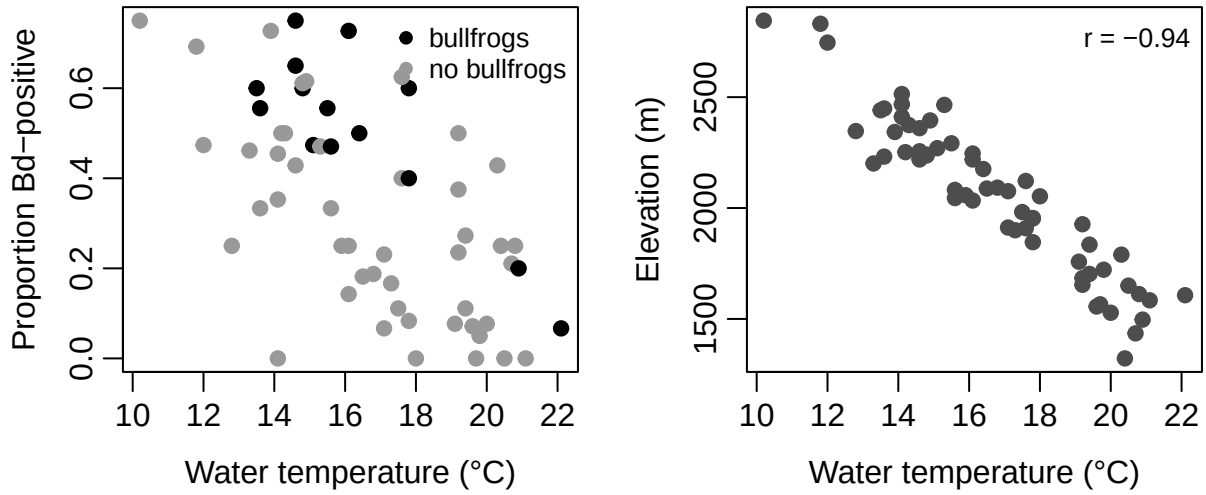
Lab 8, Act 1: Which Pond Features Matter?

Which pond features predict chytrid, and can the data tell temperature from elevation?

Answer key. Answers are in magenta sans-serif type; longer answers have a bar in the margin.

The scenario

The summer after the 12-pond survey from Lab 7, the field crew swabbed frogs at **60 ponds** across the region and recorded eight features of each pond: water temperature, elevation, canopy cover, conductivity, pond area, whether fish are present, whether bullfrogs are present (an invasive species that carries Bd without dying from it), and distance to the nearest road. Managers can afford to monitor only some ponds and want to know which features predict where Bd is common.



The model for every candidate is the logistic-binomial model from Lab 7, with more predictors: the number of Bd-positive frogs at pond i is Binomial(n_i, p_i), with

$$\text{logit}(p_i) = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \ldots$$

Every predictor was **standardized** (mean 0, SD 1) before fitting, so each β is the change in log-odds of infection per standard deviation of that predictor.

Which pond features predict Bd prevalence?

Please keep laptops closed for this Act.

Part 1: Comparing candidate models (individual, ~20 min)

The table lists seven candidate models, the number of parameters k (including β_0), and each model's negative log-likelihood (NLL) at its MLE.

1a. Fill in the missing AIC values ($\text{AIC} = 2\,\text{NLL} + 2k$) and every ΔAIC.

Model	Predictors	k	NLL	AIC	dAIC
1	intercept only	1	179.40	360.80	*139.64*
2	temp	2	137.42	*278.84*	*57.68*
3	elevation	2	137.73	*279.46*	*58.30*
4	temp + elevation	3	136.39	278.78	*57.62*
5	temp + bullfrogs + canopy	4	106.58	*221.16*	*0.00*
6	temp + elev. + bullfrogs + canopy	5	106.20	222.40	*1.24*
7	all eight	9	104.59	*227.18*	*6.02*

1b. Which model is best by AIC? Model 6 adds elevation to Model 5 and sits within 2 AIC units of it. Does that mean the data support an effect of elevation? Use the NLLs.

> Model 5 (temp + bullfrogs + canopy), AIC 221.16. No: adding elevation lowers the NLL by only 0.38. A parameter with no effect at all costs at most 2 AIC units, so landing within 2 is expected even if elevation does nothing.

1c. Use likelihood ratio tests to compare Model 5 with Model 6, and Model 5 with Model 7. For each, give the test statistic, its degrees of freedom, and your conclusion. Critical values of χ^2 at the 0.05 level: 3.84 (1 df), 5.99 (2), 7.81 (3), 9.49 (4), 11.07 (5).

> Model 5 vs. 6: $D = 2(106.58 - 106.20) = 0.76$, 1 df; $0.76 < 3.84$, so keep Model 5.
>
> Model 5 vs. 7: $D = 2(106.58 - 104.59) = 3.98$, 5 df; $3.98 < 11.07$, so keep Model 5.

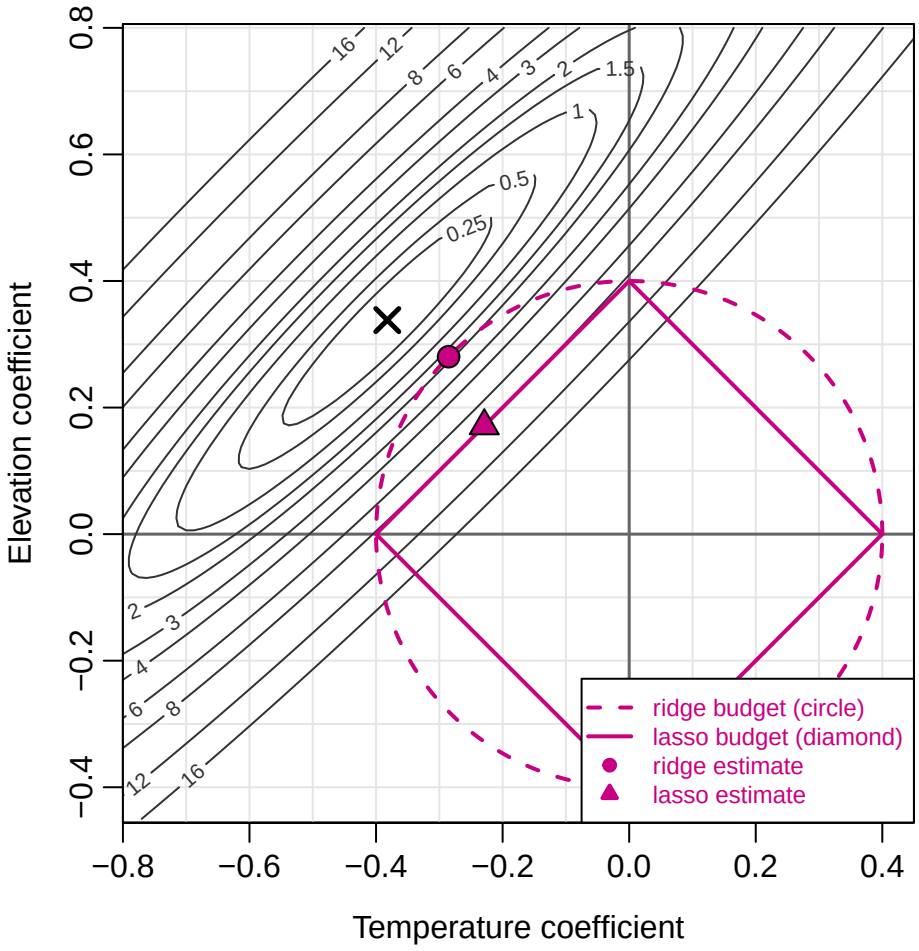
1d. Models 2 and 3 are nearly tied. Can these data tell whether water temperature or elevation is what drives Bd? Why or why not? What kind of ponds, or what kind of study, would help?

> No. Temperature and elevation are strongly correlated ($r = -0.94$), so either one alone fits about equally well (ΔAIC 0.62). Helpful: ponds where the two come apart (warm high ponds, cold low ponds, e.g. spring-fed or shaded sites), or an experiment that changes water temperature directly.

Part 2: Penalties as budgets (individual, ~15 min)

A ridge penalty, $\text{NLL}(\beta) + \lambda \sum_j \beta_j^2$, gives the same estimate as finding the lowest-NLL point inside a **budget** $\sum_j \beta_j^2 \le t^2$, for some t that shrinks as λ grows. The lasso budget is $\sum_j |\beta_j| \le t$. Neither budget includes β_0.

The figure shows NLL contours for Model 4 (temperature and elevation only), labeled by how far each lies above the minimum. The $\times$ marks the MLE.



2a. On the figure, draw the ridge budget $\beta_{temp}^2 + \beta_{elev}^2 \le 0.4^2$ and the lasso budget $|\beta_{temp}| + |\beta_{elev}| \le 0.4$. Mark the ridge estimate and the lasso estimate, and read off their approximate coordinates.

Ridge: (-0.29, 0.28) Lasso: (-0.23, 0.17) (approximate values from a sketch are fine)

2b. Did the lasso set either coefficient to exactly 0? Use the shapes in your sketch to explain why or why not.

> No. The contours are long ellipses that run parallel to the diamond's upper-left edge, so the lowest contour touches that edge partway along it, not at a corner. Both coefficients stay nonzero.

2c. What would the contours have to look like for the lasso to set the elevation coefficient to 0?

> The lowest contour would have to touch the diamond first at its corner on the temperature axis, $(-0.4, 0)$: for example, rounder contours (if temperature and elevation weren't correlated) centered near the temperature axis (a small elevation effect), or ellipses tilted across the edge rather than along it.

Group discussion and revision (~10 min)

Compare your answers with your group, especially 1b, 1d, and your sketch in 2a, and revise as needed. Record any changes you made, new insights, and any "muddiest points" or topics you want to review.

Open-ended.